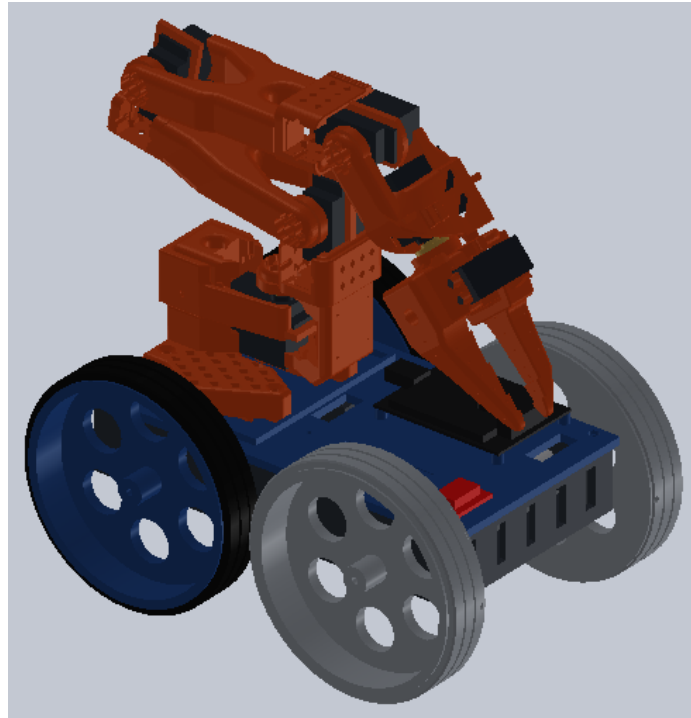


THENAR WALKER

Build book — RoboReach 2026-27 (Techfest IIT Bombay)

Every printed part with its sizes and every hole (size, position, what screw goes in it), the overall robot dimensions, where each board sits, the assembly steps, wiring, and the full parts list with what the team already has.

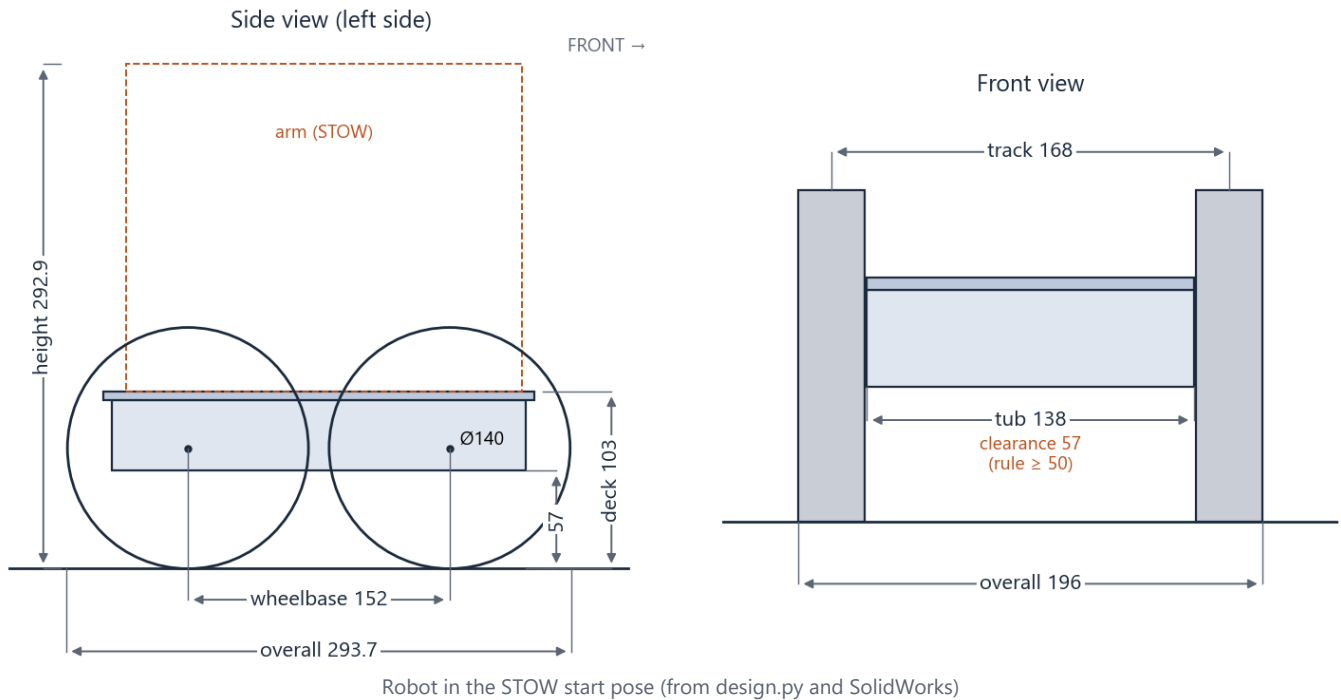


SolidWorks assembly in the STOW start pose

Section	Page
1 · Overall dimensions	2
2 · Parts and their sizes	3
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5 · Assembly, step by step	16
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Two open items change some drawings: (1) the tub motor slots are drawn for a 25 mm motor; the motors bought are 35 mm (6 mm shaft, M8 collar) — the tub will be redrawn once the motor is fully measured. (2) The deck bosses fit the Cytron MDD10A; if the SHIELD-MDD10 is used instead, they move.

1 · Overall dimensions



Item	Size	Rule / note
Overall L × W × H (STOW)	293.7 × 196.0 × 292.9	must fit 300 × 200 × 300 at the start
Ground clearance (lowest point except wheels)	57.0 (tub floor)	rule ≥ 50. With the 35 mm motor the motor bottom is ~52.5
Wheels	Ø140 × 28 wide	front: one-piece PETG; rear: PETG hub + TPU tyre
Wheelbase / track	152 / 168	axle 70 above the ground
Chassis tub	240 × 138 × 41	bottom at 57, top at 98
Deck plate	250 × 138 × 5 (+ 6 bosses, 4 fence)	top surface at 103
Arm base position	x = -87 from the centre	bolted through 4 of the arm base vent slots
Grasp height (REACH_450)	object centre at 449.6	rule: object centred at 450
Floor pick (PICK_FLOOR)	grasp centre 25.1 above the ground	50 mm cube ~254 ahead of the centre
Mass (CAD estimate)	2.40 kg	arm included
Battery voltage	11.1 V nominal, 12.6 V full	rule ≤ 24 V DC

2 · Parts and their sizes

Printed parts (PETG unless marked)

Part	Qty	Size (L × W × H)	Key holes	Drawing
RR-01 Chassis tub	1	240 × 138 × 41	4 × Ø4 insert bores, motor slots, 20 × Ø2.2 pilots	p. 4–7
RR-02 Deck plate	1	250 × 138 × 11	8 × Ø3.4, 4 × Ø2.6, 2 × Ø2.8, 2 slots, kill switch cut-out	p. 8
RR-03 Battery cage	1	128 × 44 × 27 (inside 108 × 36)	2 × Ø3.4 tabs, 4 × Ø2.6 bar pilots	p. 9
RR-04 Battery bar	2	10 × 44 × 3	2 × Ø3.4	p. 10
RR-05 Rear hub (6 mm)	2	Ø131 × 28 (rim Ø128)	Ø6.3 bore, 6 × Ø26, M3 grub	p. 11
RR-07 Front wheel (6 mm)	2	Ø140 × 28	Ø6.3 bore, 6 × Ø26, M3 grub	p. 12
RR-06F Tyre half-strip (TPU)	4	224.3 × 25 × 6	Ø4.4 key notch, click joint	p. 13

Bought parts

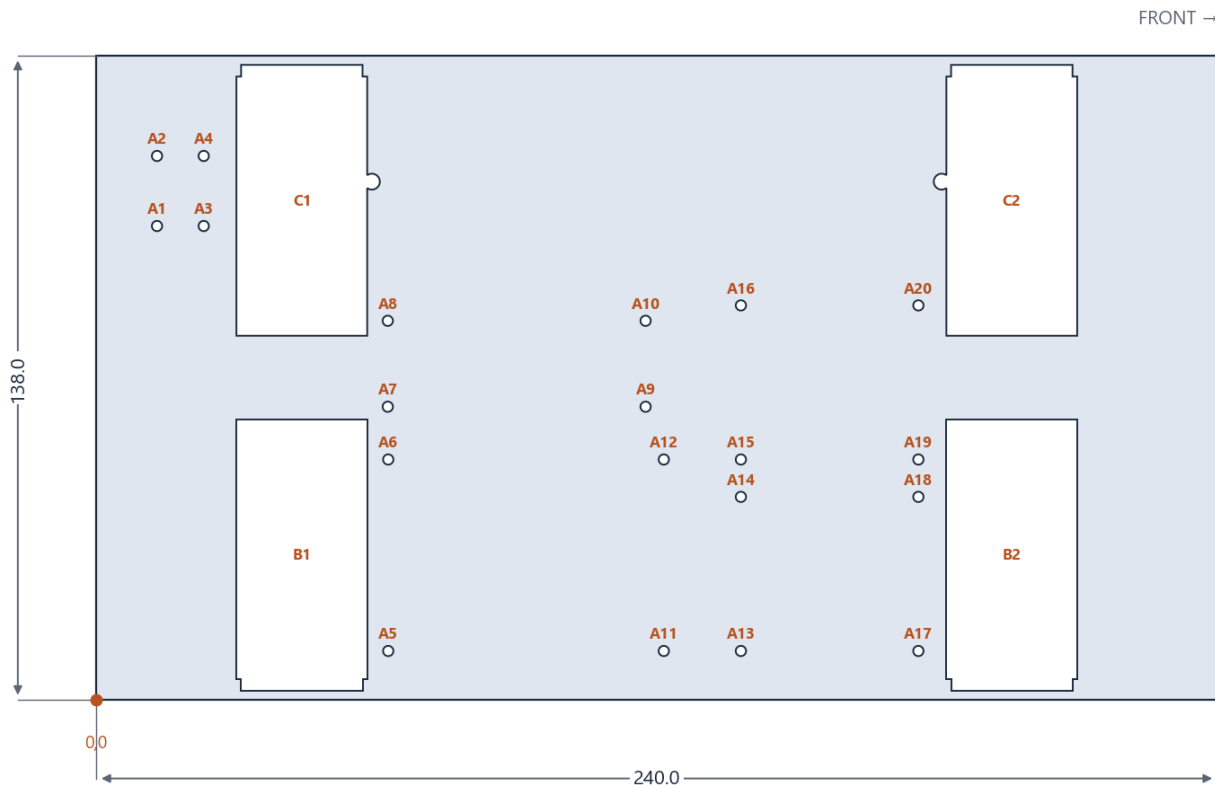
Part	Size	Mounting	Note
Gear motor 370, plastic gearbox, 12 V ~100 RPM	gearbox Ø35; shaft Ø6 round with cross-hole	M8 threaded collar + 13 mm nut	To measure: collar length, shaft length, gearbox length, can Ø × length
Cytron MDD10A Rev2.0 motor driver	84.5 × 62 × ~13	4 × Ø3 holes on 78.74 × 55.88	deck top, terminals forward
Cytron SHIELD-MDD10 (alternative)	Arduino-Uno shield size (~69 × 54)	Uno hole pattern	measure before use
ESP32-DevKitC 38-pin	55 × 28	2 × M2.5 standoffs, zip tie	under the deck
PCA9685 servo driver	62.2 × 25.4	4 × M2.5 posts	tub floor
6 V buck ≥ 10 A (XL4016 class)	66 × 48 × 18	4 × M2.5 posts	tub floor
MP1584 5 V buck	22 × 17	4 × M2.5 posts	tub rear corner
3S LiPo 2200 mAh, XT60	max 106 × 34 × 26	in the cage, 2 bars on top	
KCD4 rocker (kill switch)	cut-out 30.2 × 22.2, body 32 deep	snaps into the deck	front right
MG996R servo (arm, ×6)	40.7 × 19.7 × 42.9	arm parts	thenar-arms R3

3 · Part drawings with every hole

Each drawing is a slice through the real 3-D part. Holes are lettered by size (A = smallest); the table says what each is for, and the X / Y list gives every hole centre measured from the orange 0,0 corner (or from the centre for wheels). Check a printed part with a caliper against these numbers.

RR-01 Chassis tub — floor

On hold: the tub is drawn for the 25 mm motor. The motor slots, side-wall holes and floor windows will change for the 35 mm motor, and the battery-cage screw holes (half inside the motor windows, marked C1 / C2 notch) will move inward. Do not print plate 1 until the redesign.



Section of the real part · overall 240.0 × 138.0 mm in this view · hole centres from the 0,0 corner (X to the right, Y up)

ID	Size (mm)	Qty	What it is for	ID	X	Y	ID	X	Y	ID	X	Y
A	Ø2.2	20	Board standoff pilots (Ø6 × 3 posts): M2.5 × 6 self-tapping. 5 V buck, PCA9685, 6 V buck, 2 spare pads.	A1	13.0	101.5	A9	117.6	62.8	A17	176.0	10.5
				A2	13.0	116.5	A10	117.6	81.2	A18	176.0	43.5
				A3	23.0	101.5	A11	121.5	10.5	A19	176.0	51.5
				A4	23.0	116.5	A12	121.5	51.5	A20	176.0	84.5
				A5	62.5	10.5	A13	138.0	10.5	B1	44.0	31.1
				A6	62.5	51.5	A14	138.0	43.5	B2	196.0	31.1
				A7	62.4	62.8	A15	138.0	51.5	C1	44.1	106.9
				A8	62.4	81.2	A16	138.0	84.5	C2	195.9	106.9
B	28.0 × 58.0	2	Motor floor window (motor can drops in)									
C	30.7 × 58.0	2	Motor floor window + half of the battery-cage screw hole									

- Section through the 3 mm floor. Board standoff pilots (Ø2.2) are 1 mm deep into the floor, under 3 mm tall Ø6 posts.
- The 4 big cut-outs at the corners are the motor floor windows.

RR-01 Chassis tub — side wall

← FRONT



Section of the real part · overall 240.0 × 41.0 mm in this view · hole centres from the 0,0 corner (X to the right, Y up)

ID	Size (mm)	Qty	What it is for	ID	X	Y
A	Ø3.4	4	M3 × 6: gearbox face to wall (2 per motor, 17 apart)	A1	35.5	13.0
B	Ø8.0	2	Motor shaft + gearbox boss	A2	52.5	13.0
				A3	187.5	13.0
				A4	204.5	13.0
				B1	44.0	13.0
				B2	196.0	13.0

- Motor slots: 26 mm wide U-slots, round top centred on the axle (13 mm above the tub bottom = 70 mm above the ground), open at the bottom.
- Per motor: Ø8 shaft/boss hole + 2 × Ø3.4 (M3) holes 17 mm apart. The right wall is the mirror image.
- Sized for the 25 mm motor — will change for the 35 mm motor.

RR-01 Chassis tub — top (insert columns)

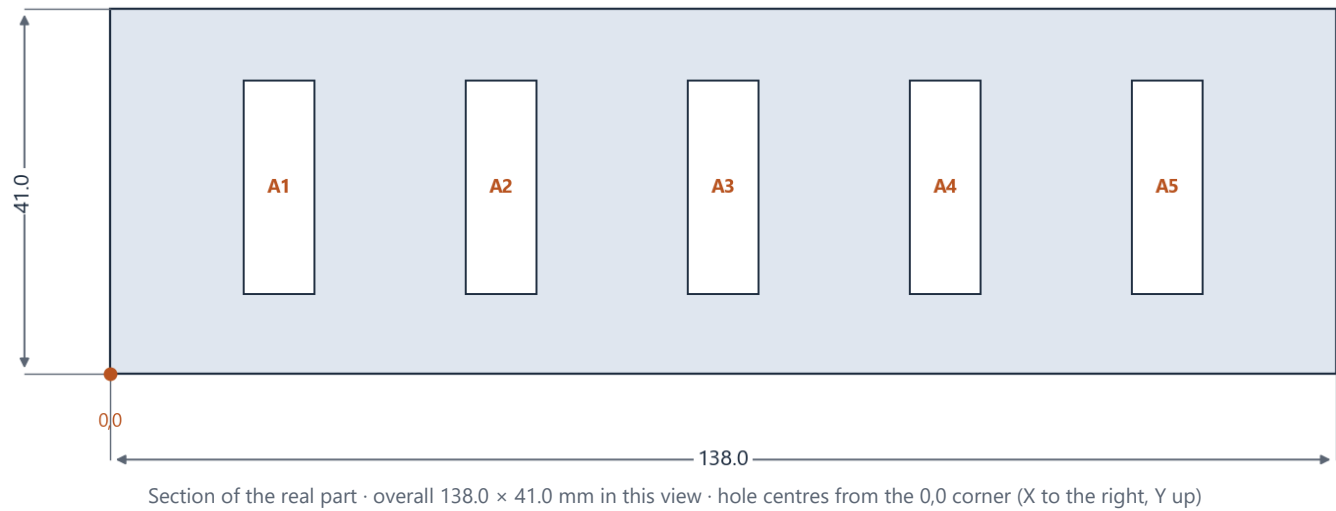


Section of the real part · overall 240.0 × 138.0 mm in this view · hole centres from the 0,0 corner (X to the right, Y up)

ID	Size (mm)	Qty	What it is for	ID	X	Y
A	Ø4.0	4	M3 heat-set insert bore, 8 deep	A1	8.0	6.0
				A2	8.0	132.0
				A3	232.0	6.0
				A4	232.0	132.0

- Corner columns Ø9 with Ø4.0 bores, 8 mm deep: melt in the M3 heat-set inserts.

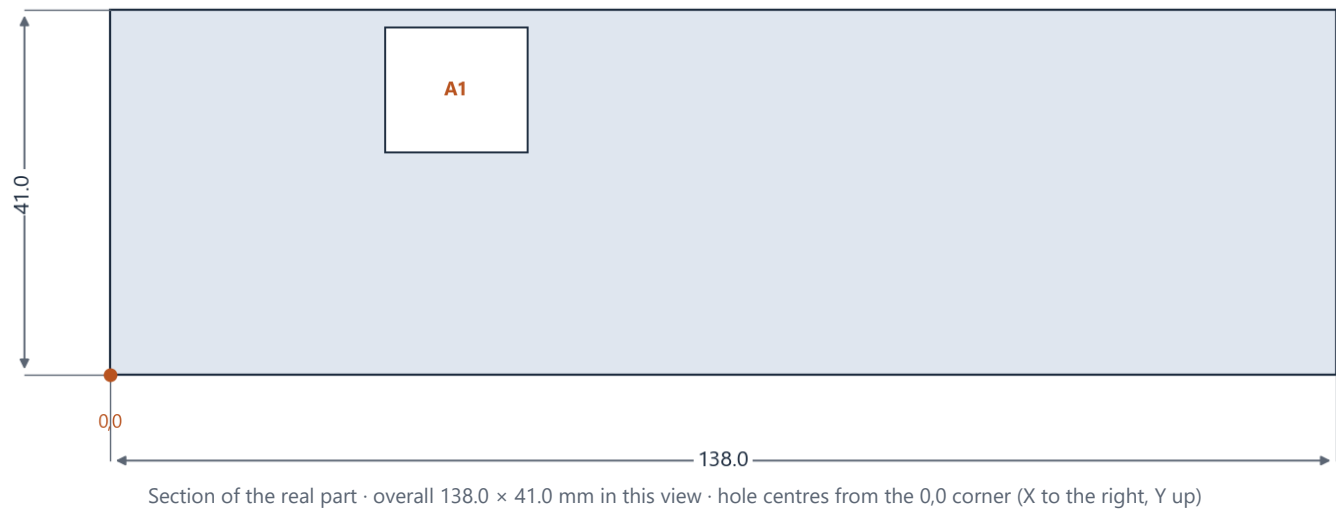
RR-01 Chassis tub — front wall



ID	Size (mm)	Qty	What it is for	ID	X	Y
A	8.0 × 24.0	5	Ventilation slot	A1	19.0	21.0
				A2	44.0	21.0
				A3	69.0	21.0
				A4	94.0	21.0
				A5	119.0	21.0

- 5 ventilation slots 8 × 24 mm.

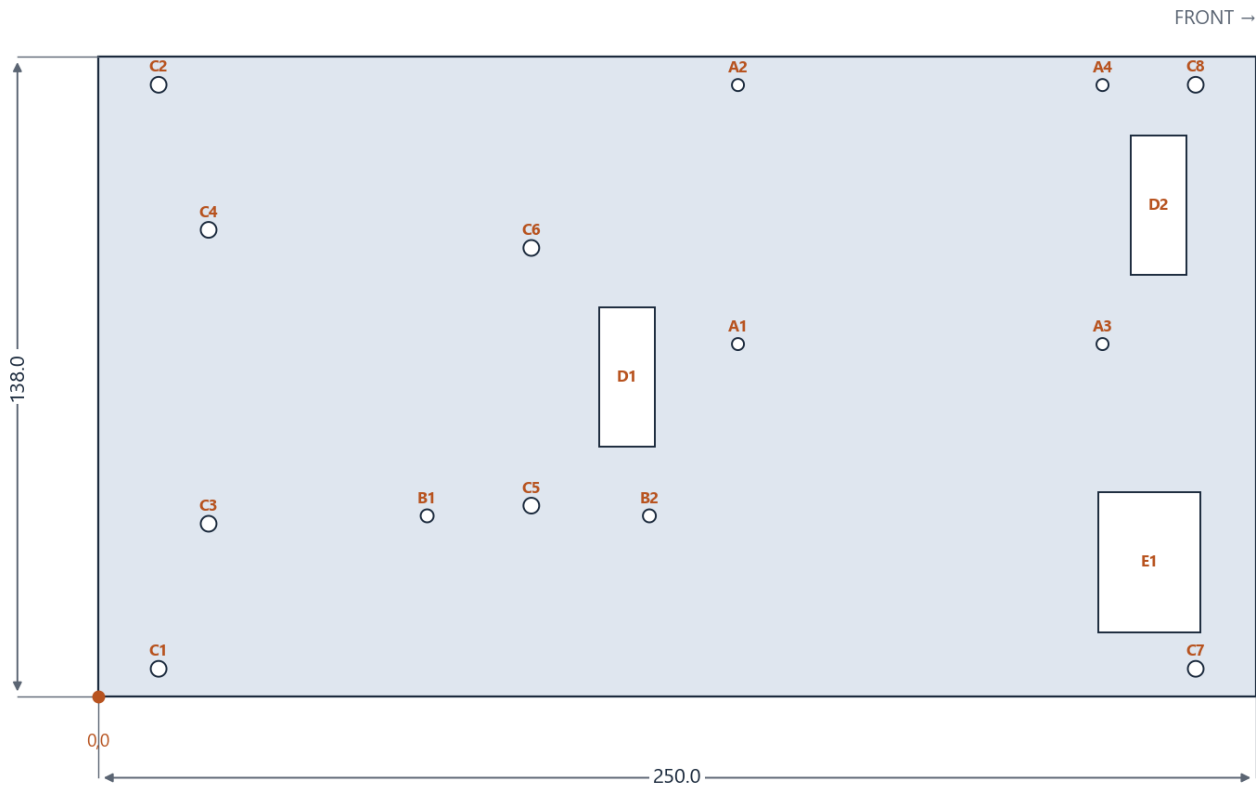
RR-01 Chassis tub — rear wall



ID	Size (mm)	Qty	What it is for	ID	X	Y
A	14.0 × 16.0	1	ESP32 USB cable window	A1	39.0	32.0

- USB grommet window 16 × 14 mm for the ESP32 cable.

RR-02 Deck plate



Section of the real part · overall 250.0 × 138.0 mm in this view · hole centres from the 0,0 corner (X to the right, Y up)

ID	Size (mm)	Qty	What it is for	ID	X	Y	ID	X	Y	ID	X	Y
A	Ø2.6	4	Motor driver MDD10A: M3 × 10 self-tapping (pilot in a Ø7 × 6 mm boss)	A1	138.1	76.1	C1	13.0	6.0	C7	237.0	6.0
				A2	138.1	131.9	C2	13.0	132.0	C8	237.0	132.0
				A3	216.9	76.1	C3	23.8	37.3	D1	114.2	69.0
				A4	216.9	131.9	C4	23.8	100.7	D2	229.0	106.0
B	Ø2.8	2	ESP32: M2.5 × 10 brass standoffs (hangs under the deck)	B1	71.0	39.0	C5	93.5	41.2	E1	227.0	29.0
				B2	119.0	39.0	C6	93.5	96.8			
C	Ø3.4	8	4 corners: M3 × 10 deck → tub inserts (counterbored). Inner 4: arm base M3 × 20 + washer + nyloc									
D	12.0 × 30.0	2	Slots: servo + signal wires (by the arm), motor + power wires (in front of the driver)									
E	22.2 × 30.2	1	Kill switch cut-out (KCD4 snaps in)									

- Ø3.4 corner holes, counterbored Ø6.5 × 2.5 deep on top: deck → tub, M3 × 10.
- Arm base: 4 × Ø3.4 through the arm vent slots, M3 × 20 + nyloc under the deck.
- MDD10A: 4 × Ø2.6 pilots in Ø7 × 6 mm bosses on top, 78.74 × 55.88 apart (M3 × 10 self-tapping).
- ESP32: 2 × Ø2.8 for the M2.5 brass standoffs.

RR-03 Battery cage — base

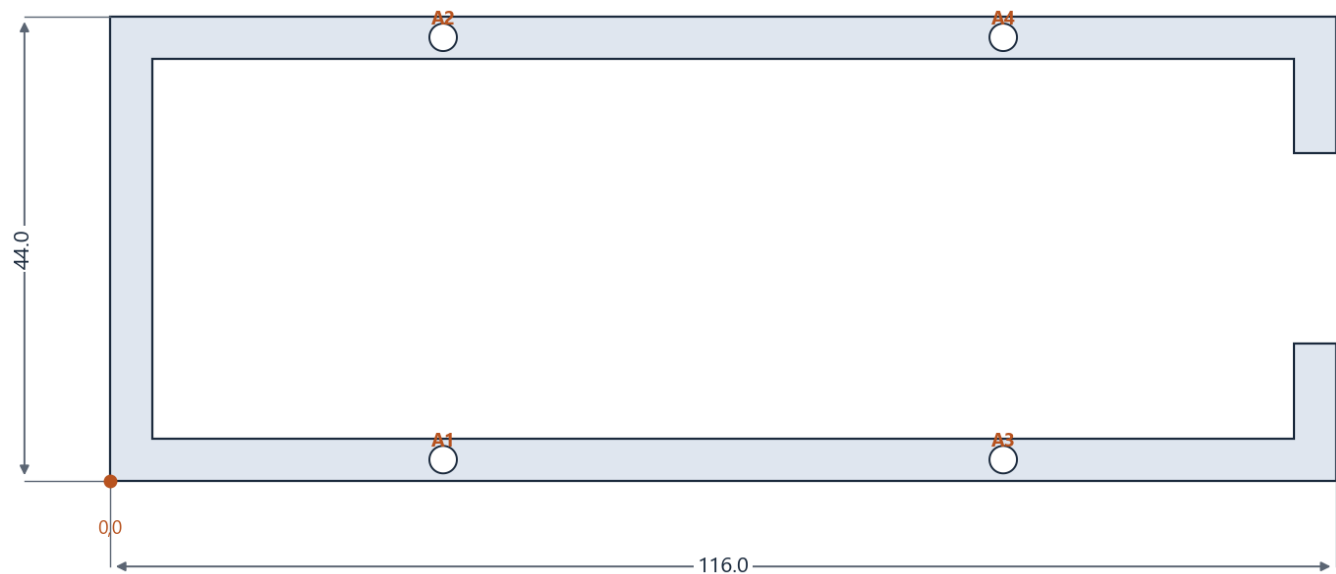


Section of the real part · overall 128.0 × 44.0 mm in this view · hole centres from the 0,0 corner (X to the right, Y up)

ID	Size (mm)	Qty	What it is for	ID	X	Y
A	Ø3.4	2	M3 × 8 + nyloc through the tub floor	A1	3.0	22.0
				A2	125.0	22.0

- Inside 108 × 36 mm (battery max 106 × 34 × 26). Tabs: 2 × Ø3.4 for M3 × 8 + nyloc through the tub floor.

RR-03 Battery cage — top

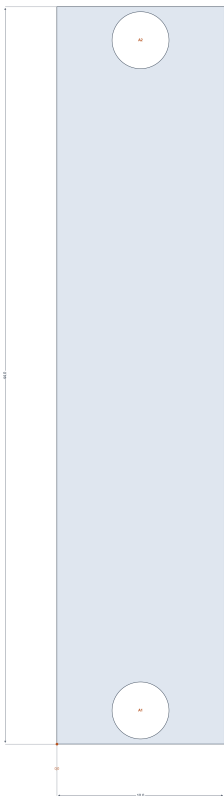


Section of the real part · overall 116.0 × 44.0 mm in this view · hole centres from the 0,0 corner (X to the right, Y up)

ID	Size (mm)	Qty	What it is for	ID	X	Y
A	Ø2.6	4	M3 × 8 self-tapping for the bars	A1	31.5	2.0
				A2	31.5	42.0
				A3	84.5	2.0
				A4	84.5	42.0

- 4 × Ø2.6 pilots, 8 mm deep: M3 × 8 self-tapping for the two bars.

RR-04 Battery bar (× 2)



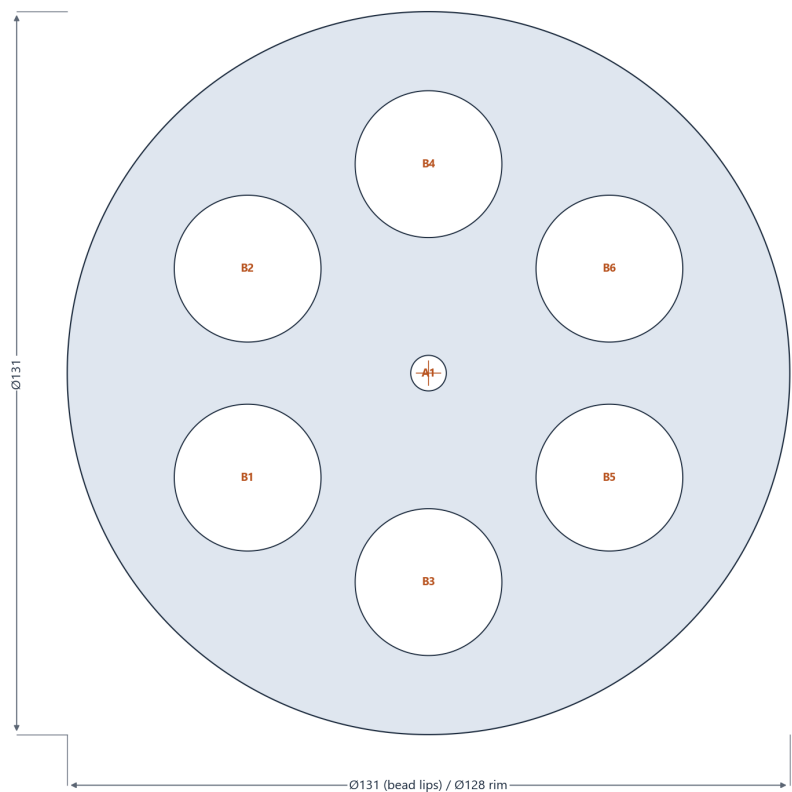
Section of the real part · overall 10.0 × 44.0 mm in this view · hole centres from the 0,0 corner (X to the right, Y up)

ID	Size (mm)	Qty	What it is for
A	Ø3.4	2	M3 × 8 clearance (into the cage bosses)

ID	X	Y
A1	5.0	2.0
A2	5.0	42.0

• 3 mm thick.

RR-05 Rear wheel hub (6 mm shaft)

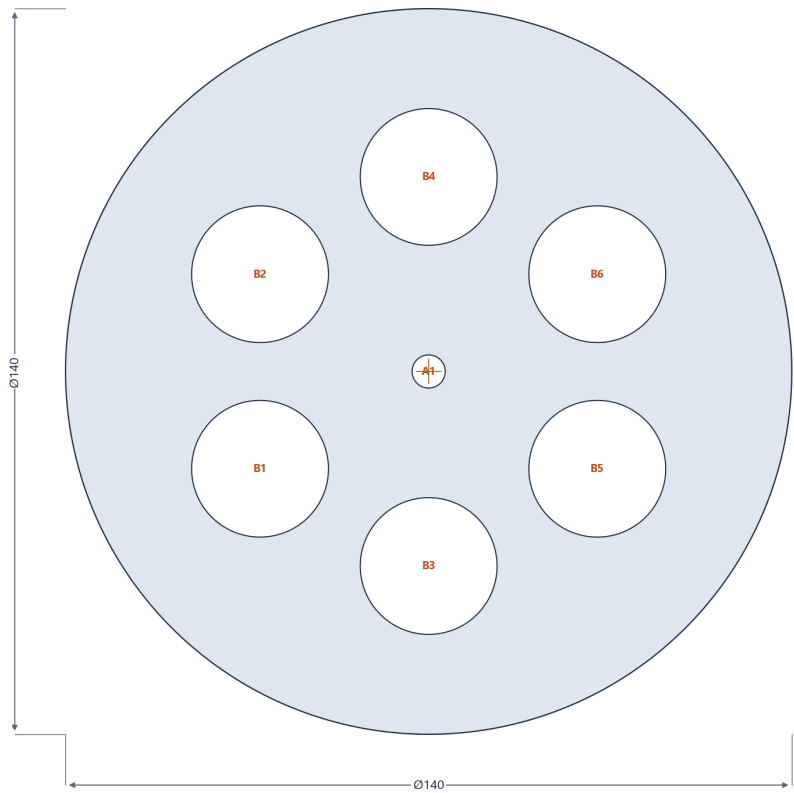


Section of the real part · overall 128.0 × 128.0 mm in this view · hole centres from the wheel centre (X to the right, Y up)

ID	Size (mm)	Qty	What it is for	ID	X	Y
A	Ø6.3	1	Motor shaft (6 mm round)	A1	0.0	-0.0
B	Ø26.0	6	Lightening holes	B1	-32.0	-18.5
				B2	-32.0	18.5
				B3	0.0	-37.0
				B4	-0.0	37.0
				B5	32.0	-18.5
				B6	32.0	18.5

- Positions from the wheel centre. Rim Ø128 between two bead lips Ø131; width 28.
- Bore Ø6.3 round. M3 grub screw: Ø2.6 radial hole 3.6 mm from the inner face, hex-key access Ø3.8 through the rim.

RR-07 Front wheel (6 mm shaft)

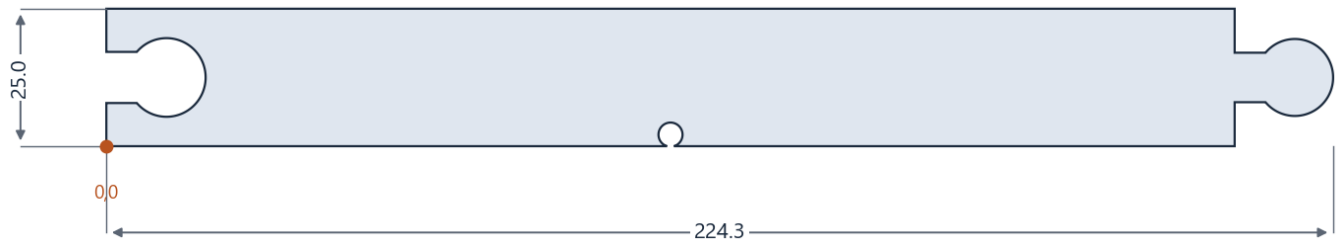


Section of the real part · overall 138.0 × 138.0 mm in this view · hole centres from the wheel centre (X to the right, Y up)

ID	Size (mm)	Qty	What it is for
A	Ø6.3	1	Motor shaft (6 mm round)
B	Ø26.0	6	Lightening holes

ID	X	Y
A1	0.0	-0.0
B1	-32.0	-18.5
B2	-32.0	18.5
B3	0.0	-37.0
B4	-0.0	37.0
B5	32.0	-18.5
B6	32.0	18.5

- Positions from the wheel centre. Ø140 one-piece, width 28, tread with 2 grooves.
- Bore Ø6.3 round. Grub screw as the rear hub; the key access hole goes through the tread.

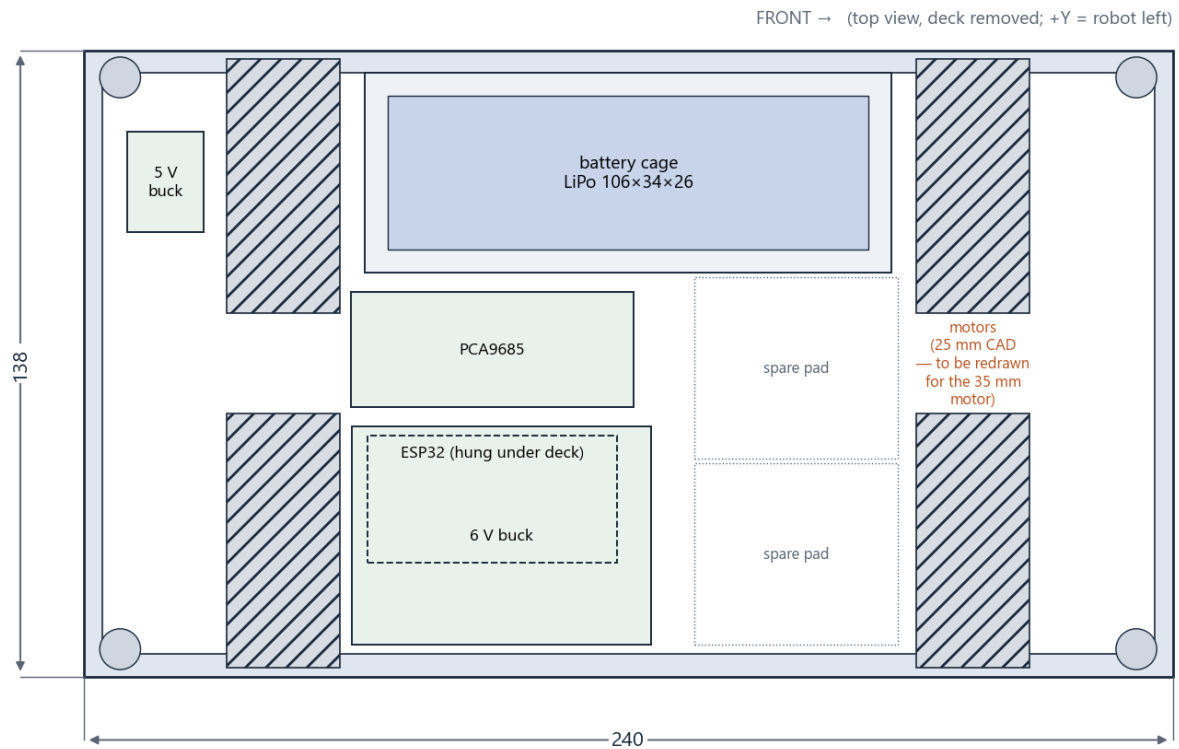
RR-06F TPU tyre half-strip

Section of the real part · overall 224.3 × 25.0 mm in this view · hole centres from the 0,0 corner (X to the right, Y up)

- 6 mm thick, 25 mm wide, 224.3 mm long including the knob. Two halves make one tyre (412.6 mm, 2 % shorter than the hub).
- Ø4.4 key hole sits over the hub grub screw (edge with the hole toward the robot).

4 · Layout: where everything sits

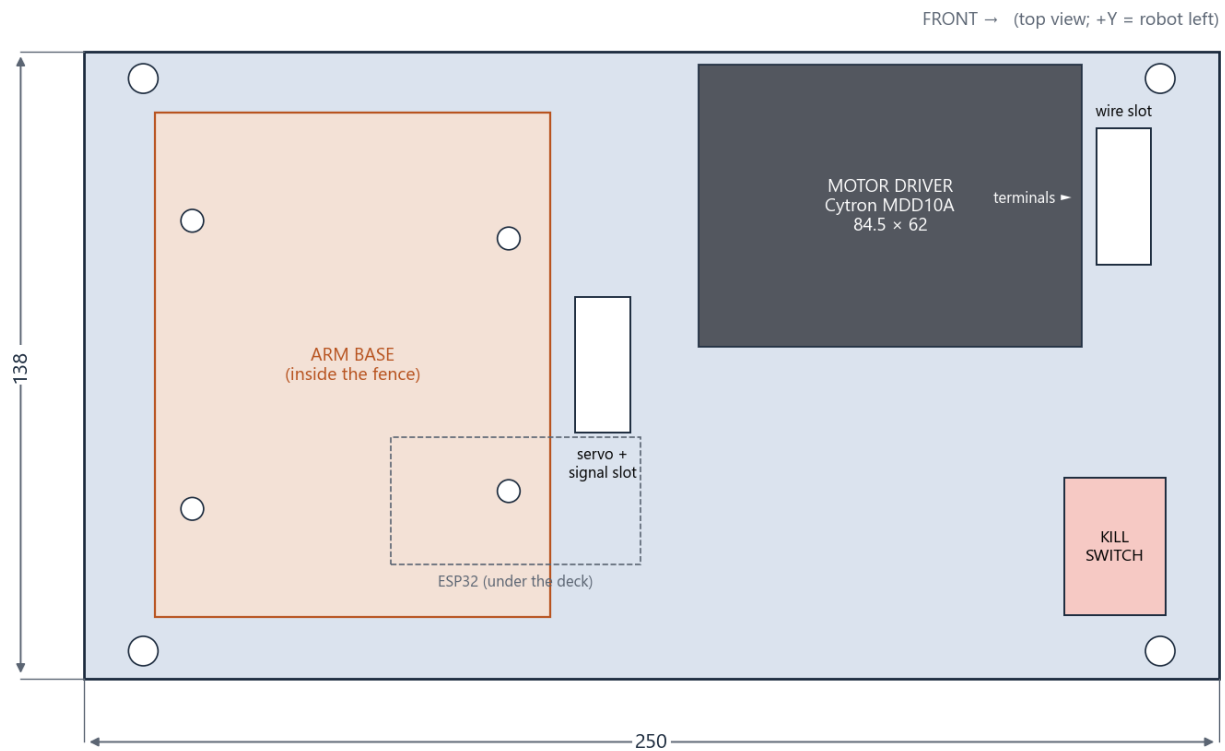
Tub floor (deck removed)



Top view. Positions are board centres from the tub centre; +X forward, +Y to the robot left.

Item	Centre X	Centre Y	Size	Mounted on
6 V buck	-28.0	-38.0	66 × 48	floor posts (3 mm)
Motor driver MDD10A	52.5	35.0	84.5 × 62	deck top (6 mm bosses)
PCA9685	-30.0	3.0	62.2 × 25.4	floor posts (3 mm)
5 V buck (MP1584)	-102.0	40.0	22 × 17 (turned 90°)	floor posts (3 mm)
ESP32	-30.0	-30.0	55 × 28	under the deck (10 mm standoffs)
Battery (in the cage)	0.0	42.0	106 × 34	cage on the floor
Kill switch	102.0	-40.0	30.2 × 22.2 cut-out	deck
Arm base origin	-87.0	0.0	fence inside 88 × 112	deck

Deck top



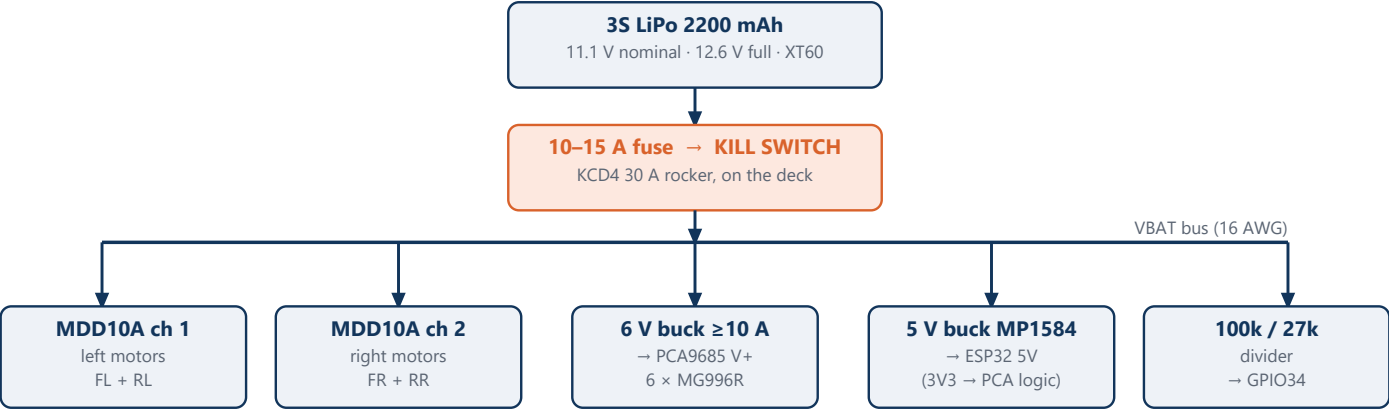
Top view of the deck. The driver terminals face forward over the wire slot.

Motor wheels: front-left, rear-left, front-right, rear-right motors sit at $X = \pm 76$ (wheelbase 152), axle 70 above the ground, shafts pointing outward through the side walls.

5 · Assembly, step by step

0 · Before you start	Caliper, the motor, the drivers
- Measure one motor: collar length, shaft length past the collar, gearbox length, can Ø × length. The tub is redrawn from these. - Decide the driver: MDD10A (deck ready) or SHIELD-MDD10 (deck bosses move). - Bottle test done: one motor lifted 750 ml at 7 cm ($\geq \sim 5.2$ kg·cm), enough for the course.	
1 · Print	P1S, ~0.8 kg PETG, ~85 g TPU per tyre plate
- PETG: plate 1 (tub, cage, bars — after the redesign), plate 2 (deck), plates 3 + 4b (wheels). TPU: plate 5F (tyre strips) × 1, × 2 for spares. 0.2 mm layers, 4 walls, 30 % gyroid (TPU: 3 walls, 20 %), no supports. TPU from the external spool, not the AMS.	
2 · Clean up the prints	6 mm drill, M3 tap (optional)
- Run a 6 mm drill through each wheel bore by hand; drill the 2 older 4 mm-bore wheels out to 6 mm. - Tap or self-thread the Ø2.6 grub holes with an M3 grub screw. - Check holes against the drawings (section 3) with a caliper.	
3 · Tub	4 inserts, 2 × M3×8 + nyloc, ~12 × M2.5×6, motor mounts
- Melt 4 × M3 heat-set inserts into the corner bores (Ø4.0, 8 deep). - Screw the battery cage to the floor: 2 × M3 × 8 + nyloc from below. - Mount the 4 motors in the side walls, shafts out (new mount for the 35 mm motor). Solder leads first. - Screw the 6 V buck, PCA9685 and 5 V buck to their floor posts (M2.5 × 6 self-tapping). - Set the 6 V buck to 6.0 V and the 5 V buck to 5.0 V with a multimeter, before connecting anything.	
4 · Wheels and tyres	4 × M3×5 grub, superglue
- Rear: click two TPU half-strips together, superglue the joint, wrap around a rear hub between the lips (grooves out, notch over the grub screw), pull the second joint closed and glue. - Push each wheel onto its shaft ~1 mm off the wall; turn it so the grub screw lines up with the shaft cross-hole; tighten the M3 × 5 grub. - Front wheels (hard, one piece) at the front, TPU-tyred wheels at the rear. Spin each wheel by hand: no rubbing.	
5 · Deck	2 standoffs, 4 × M3×10 self-tap, zip ties
- Screw 2 × M2.5 × 10 brass standoffs under the deck, zip-tie the ESP32 to them (USB toward the rear window). - Snap the kill switch into its cut-out (front right) and wire it in series with battery +. - Motor driver on the 4 bosses (M3 × 10 self-tapping), blue terminals facing forward. - Route: servo leads + 5 driver signal wires up through the slot by the arm; motor + power wires up through the slot in front of the driver.	
6 · Close up and fit the arm	4 × M3×8 self-tap, 4 × M3×10, 4 × M3×20 + nyloc
- Battery into the cage, lead out of the front window, screw on the two bars (M3 × 8 self-tapping). - Deck onto the tub: 4 × M3 × 10 into the inserts. - Arm base inside the fence, 4 × M3 × 20 + washers through the base vent slots, nyloc nuts under the deck. - Servos J1...J6 into PCA9685 channels 0...5.	
7 · Firmware and first power-up	USB cable, multimeter
- python firmware/tools/new_key.py "Team Name" → flash thenar_walker_bot (robot) and thenar_walker_controller (leader). - Wheels off the ground, kill switch in reach. Use the driver test buttons (M1A/M1B/M2A/M2B) first. - Joystick check: if a whole side runs backwards swap that side's two driver wires. - Calibrate the servos (calibration.h), set FOLLOWER_CALIBRATED = true, reflash. - Failsafe test: switch the controller off while holding a cube → wheels stop, arm freezes, gripper holds.	

6 · Wiring



All grounds common. Every voltage on the robot ≤ 12.6 V (rule: ≤ 24 V DC).
Kill switch is before everything → it shuts the whole robot down (rule).

Motors: both left motors in parallel on channel 1, both right motors on channel 2 — same wire of each motor into the same terminal.

ESP32 pin	Signal	MDD10A (white 5-pin plug)	SHIELD-MDD10 (jumper-selected)
GPIO25	left speed	PWM1 (pin 4)	PWM1 pin, e.g. D3
GPIO26	left direction	DIR1 (pin 5)	DIR1 pin, e.g. D2
GPIO16	right speed	PWM2 (pin 2)	PWM2 pin, e.g. D6
GPIO17	right direction	DIR2 (pin 3)	DIR2 pin, e.g. D7
GND	ground	GND (pin 1)	any GND
GPIO21 / 22	I ² C	PCA9685 SDA / SCL	
GPIO13	servo enable	PCA9685 OE	
GPIO34	battery sense	100 kΩ / 27 kΩ divider	
GPIO2	status LED	on-board	solid = ACTIVE

10 kΩ from PWM1 to GND and from PWM2 to GND (no twitch while the ESP32 boots). Fuse 10 A, close to the XT60. Kill switch before everything. Never power the servos from the ESP32 or USB.

7 · Bill of materials and status

Have = bought / built. To buy = still needed. Confirm = not reported yet. Decide = two options, only one needed.

Part	Qty	Spec	Status
370 gear motor, plastic gearbox, 12 V ~100 RPM	4 (+1)	Ø35 gearbox, 6 mm shaft, M8 collar; bottle test passed	Have
M3 × 5 cup-point grub screw	4	wheel → shaft	To buy
22 AWG silicone wire red + black	2 m each	motor leads	Confirm
Cytron MDD10A Rev2.0	1	dual 10 A, PWM + DIR	Decide · have
Cytron SHIELD-MDD10	1	same driver as an Arduino shield	Decide · have
10 kΩ resistor	2	PWM pull-downs	To buy
3S LiPo 2200 mAh 30C XT60	1 (+1)	max 106 × 34 × 26	Confirm
LiPo balance charger + safe bag	1	3S balance	Confirm
XT60 pair	2		Have
Glass fuse holder 5 × 20 + 10 A fuse	1	inline at the XT60	Have
LiPo cell tester / buzzer	1	on the balance lead	Have
KCD4 rocker 16 A+ (kill switch)	1	cut-out 30.2 × 22.2	Confirm
6 V buck ≥ 10 A	1	servo rail	Confirm
MP1584 5 V buck	1	ESP32 supply	Confirm
16 AWG silicone wire red + black	1 m each	power	Confirm
100 kΩ + 27 kΩ resistors	1 each	battery sense	To buy
ESP32-DevKitC 38-pin	1	robot	Confirm
PCA9685	1	servo driver	Confirm
Jumper wires, servo extensions, heat-shrink, zip ties	1 lot		Confirm
thenar-arms follower R3 (6 × MG996R)	1	the arm	Have
thenar-arms L1 leader (controller)	1		Have
Joystick module, ENABLE toggle, SPEED button, power bank	1 each	controller add-ons	Confirm
M3 heat-set inserts	4	tub corners	Confirm
M3 × 10 socket screws	4	deck → tub	Confirm
M3 × 10 self-tapping screws	4	driver → deck	To buy
M3 × 8 screws + nyloc	2 + 2	cage → floor	Confirm
M3 × 8 self-tapping screws	4	bars → cage	Confirm
M3 × 20 screws + washers + nyloc	4	arm → deck	Confirm
M2.5 × 10 brass standoffs + M2.5 × 6 screws	2 + 2	ESP32	Confirm
M2.5 × 6 self-tapping screws	~12	boards → floor	Confirm
Superglue (CA)	1	tyre strip joints	To buy
Plate 1: tub + cage + bars	1	PETG	On hold
Plate 2: deck	1	PETG	Ready
Plate 3: front wheel + rear hub	1	PETG (drill to 6 mm)	Printed
Plate 4b: front wheel + rear hub, 6 mm	1	PETG	Printed
Plate 5F: 4 TPU strips = 2 tyres	1 (+1 spare)	TPU 95A	Printing
25GA-370 motor, MDD3A, TB6612, white plastic wheel	–	earlier plan / unsuitable	Not used

8 · Print plates and checklist

Plate	Parts	Material	Time	Filament	Status
1	tub, cage, 2 bars	PETG	~7.1 h	~259 g	On hold
2	deck	PETG	3 h 27 min	117 g	Ready
3	front wheel + rear hub (4 mm)	PETG	6 h 08 min	202 g	Printed
4b	front wheel + rear hub (6 mm)	PETG	6 h 07 min	202 g	Printed
5F	4 tyre strips = 2 tyres	TPU 95A	5 h 40 min	84 g	Printing

PETG: 0.2 mm, 4 walls, 30 % gyroid, textured PEI, no supports. TPU: 3 walls, 20 %, external spool, dried. Sliced files: [cad/print/sliced/](#).

Before every run	
Battery $\geq$ 11.4 V (controller shows batt)	☐
Kill switch reachable and working	☐
team_config.h generated (not the example)	☐
Arm folded to STOW, robot inside 300 × 200 × 300	☐
All 4 grub screws tight, tyre joints intact	☐
Failsafe test: controller off → robot stops, gripper holds	☐